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● Concepts of Clustering and Classification:

Feature	Classification (Supervised Learning)	Clustering (Unsupervised Learning)
Learning Type	Supervised	Unsupervised
Data Labels	Has pre-existing labels (Ground truth is known)	No pre-existing labels (Ground truth is unknown)
Primary Goal	Predict the target label/category for new, unseen data	Discover hidden patterns or natural groupings inherent in the data
How It Works	Trains on labeled past data to learn boundaries between categories	Measures mathematical distance/similarity between data points to form groups
Number of Groups	Fixed and defined before training (e.g., 2 classes: Spam vs. Not Spam)	Determined by algorithm or chosen via validation metrics (e.g., Elbow Method)
Real-World Examples	• Email Spam Detection • Disease Diagnosis (Positive / Negative) • Credit Approval (Approve / Reject)	• Mall Customer Segmentation • Document Categorization • Anomaly & Fraud Detection